Propulsion & Station-Keeping: System Design Description

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Author	Jory Aster
Approved by	Cael Merrow
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1. System overview

Propulsion keeps the station in its assigned orbit and attitude. Station-keeping thrusters and sublight ion engines are commanded by the navigation computer, with attitude control gyroscopes for fine pointing.

The system is led by Propulsion & Station-Keeping. It is delivered through 6 work packages with a combined budget of 83,136.67 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
PR-SKT	Station-keeping thruster	SHV-PR-SPEC-0101	SHV-PR-DWG-0201	SHV-PR-BOM-0301
PR-SIE	Sublight ion engine	SHV-PR-SPEC-0102	SHV-PR-DWG-0202	SHV-PR-BOM-0302
PR-NAV	Navigation computer	SHV-PR-SPEC-0103	SHV-PR-DWG-0203	SHV-PR-BOM-0303
PR-ACG	Attitude control gyroscope	SHV-PR-SPEC-0104	SHV-PR-DWG-0204	SHV-PR-BOM-0304

3. Architecture

3.1 Station-keeping thruster

The station-keeping thruster is specified in SHV-PR-SPEC-0101. Its governing parameter is the thrust, set at 1,579 kN. The responsible engineer is Jory Aster.

3.2 Sublight ion engine

The sublight ion engine is specified in SHV-PR-SPEC-0102. Its governing parameter is the engine count, set at 33 ea. The responsible engineer is Tarn Falkner.

3.3 Navigation computer

The navigation computer is specified in SHV-PR-SPEC-0103. Its governing parameter is the orbit solution time, set at 49 s. The responsible engineer is Gessa Greaves.

3.4 Attitude control gyroscope

The attitude control gyroscope is specified in SHV-PR-SPEC-0104. Its governing parameter is the angular momentum, set at 5.1 GN·m·s. The responsible engineer is Emrys Stagg.

4. Interfaces

Document	Title
SHV-ICD-PD-PR	Interface Control Document: Power Distribution / Propulsion & Station-Keeping

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-PR-01	Propulsion & Station-Keeping: Design & analysis	Soren Mordant	IN_PROGRESS	78.3%
WP-SHV-PR-02	Propulsion & Station-Keeping: Fabricate & install station-keeping thruster	Tilda Falkner	IN_PROGRESS	77.8%
WP-SHV-PR-03	Propulsion & Station-Keeping: Fabricate & install sublight ion engine	Pela Illing	IN_PROGRESS	79.6%
WP-SHV-PR-04	Propulsion & Station-Keeping: Fabricate & install navigation computer	Edric Kade	IN_PROGRESS	85.8%
WP-SHV-PR-05	Propulsion & Station-Keeping: Fabricate & install attitude control gyroscope	Edda Ingram	COMPLETE	100.0%
WP-SHV-PR-06	Propulsion & Station-Keeping: Integration & test	Edric Kade	IN_PROGRESS	52.8%

6. Risks

- Supply of station-keeping thruster assemblies depends on a single supplier.
- Supply of sublight ion engine assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.